

Backward Uniqueness and Uniqueness of Asymptotically Conical Shrinkers

Kyeongho Bang

March 12, 2025

1 Overview

In these two lectures, I will present Lu Wang's 2014 paper, *Uniqueness of Self-Similar Shrinkers with Asymptotically Conical Ends*. Following is the main result of this paper:

Theorem 1.1 (L. Wang, 2014). *Let $C \subset \mathbb{R}^{n+1}$ be a regular cone with vertex O and R_0 a positive constant. Suppose that Σ and $\tilde{\Sigma}$ are noncompact, smooth, connected, properly embedded self-shrinkers in $\mathbb{R}^{n+1} \setminus B_{R_0}$ with their boundaries in S_{R_0} . If Σ and $\tilde{\Sigma}$ are asymptotic to the same cone C , i.e., $\lambda\Sigma$ and $\lambda\tilde{\Sigma}$ converge to C locally smoothly as $\lambda \rightarrow 0^+$, then Σ coincide with $\tilde{\Sigma}$.*

- Just like any other uniqueness problem in geometric PDE, one wishes to reformulate the geometric problem into a PDE, and then apply relevant uniqueness theorem on that PDE to obtain the result.
- The usual trick in this type of uniqueness problem is to use the (elliptic) strong unique continuation principle. Nevertheless, even in the simplest scenario, the conditions are not met to use the well-known elliptic unique continuation theorems directly due to strong degeneracy of the equation.
- Wang resolves this issue by employing the “backward uniqueness” machinery introduced in Escobar-Seregin-Šverák's 2003 paper *Backward Uniqueness for Parabolic Equations*.

Related works:

- Involving Carleman estimate – Kotschwar-Wang 2015: *Rigidity of asymptotically conical shrinking gradient Ricci solitons*; Wang 2016: *Uniqueness of self-similar shrinkers with asymptotically cylindrical ends*;
- An elliptic proof of [Theorem 1.1](#) – J. Bernstein 2017: *Asymptotic structure of almost eigenfunctions of drift Laplacians on conical ends*;
- An elliptic proof of a similar uniqueness result in RF – Deruelle-Schulze 2020: *A relative entropy and a unique continuation result for Ricci expanders*.

2 Backward Uniqueness and Carleman Estimates (week 1)

In the first part of this lecture, we will focus on Escauriaza-Seregin-Šverák's **backward uniqueness for parabolic equations**. This uniqueness principle is now considered a standard ingredient in the regularity theory of Navier-Stokes equation.

Theorem 2.1 (Backward uniqueness, Escauriaza-Seregin-Šverák, 2003). *Assume that $u \rightarrow Q_{R,T} := \mathbb{R}^n \setminus B_R \times [0, T]$ satisfies the inequalities*

$$|\Delta u + \partial_t u| \leq M (|u| + |\nabla u|) \text{ and } |u(x, t)| \leq M e^{M|x|^2} \text{ in } Q_{R,T} \quad (1)$$

for some positive constants M, T and R . Then, if $u(x, 0) \equiv 0$ in $\mathbb{R}^n \setminus B_R$, u vanishes identically in $Q_{R,T}$.

2.1 Unique Continuation for Elliptic Operators

First we briefly introduce Garofalo-Lin's **strong unique continuation principle for elliptic operators**. This is considered as a standard technique not only in uniqueness problems in geometric PDEs but also in the elliptic PDE theory as a whole; it plays a fundamental role for many other uniqueness principles in PDE.

In an open, connected subset $\Omega \subset \mathbb{R}^n$, $n \geq 3$, we consider a second-order elliptic equation

$$Lu = -\operatorname{div}(A(x)\nabla u) + \mathbf{b}(x) \cdot \nabla u + V(x)u = 0, \quad (2)$$

where the coefficients of L satisfy the following:

1. (uniform ellipticity) there exists a $\lambda \in (0, 1)$ such that, for every $x \in \Omega$ and $\xi \in \mathbb{R}^n$,

$$\lambda |\xi|^2 \leq \sum_{i,j=1}^n a_{ij}(x) \xi_i \xi_j \leq \lambda^{-1} |\xi|^2; \quad (3)$$

2. (uniformly Lipschitz) there exists $\Gamma > 0$ such that, for every $x, y \in \Omega$ and $i, j = 1, \dots, n$,

$$|a_{ij}(x) - a_{ij}(y)| \leq \Gamma |x - y|; \quad (4)$$

3. for every $x_0 \in \Omega$, there exists $r_0 > 0$, $f : (0, r_0) \rightarrow \mathbb{R}^+$ and $C > 0$ such that f is decreasing on $(0, r_0)$,

$$\int_0^{r_0} \frac{dr}{r f(r)} < \infty, \quad (5)$$

and for every $x \in \Omega \cap B_{r_0}(x_0)$,

$$|\mathbf{b}(x)| \leq \frac{C}{|x - x_0| f(|x - x_0|)}, \quad |V(x)| \leq \frac{C}{|x - x_0|^2 f(|x - x_0|)}. \quad (6)$$

Definition 2.2 (Unique continuation property). *A function $u \in L^2_{loc}(\mathbb{R}^n)$ is said to have the **vanish***

of infinite order at a point $x_0 \in \mathbb{R}^n$ if

$$\int_{|x-x_0|<R} u^2 dx = O(R^k) \text{ for every } k \in \mathbb{N} \text{ as } R \rightarrow 0. \quad (7)$$

Given L as in (2), L is said to have the **strong unique continuation property** in Ω if the only $H_{loc}^{1,2}$ solution of $Lu = 0$ which vanishes of infinite order at a point $x_0 \in \Omega$ is $u \equiv 0$. L is said to have the **unique continuation property** in Ω if the only $H_{loc}^{1,2}(\Omega)$ solution of $Lu = 0$ which can vanish in an open subset of Ω is $u \equiv 0$.

Theorem 2.3 (Garofalo-Lin, 1987). *Let L , as in (2), satisfy (3)-(6). Then L has the strong unique continuation property in Ω .*

Key idea of the proof is to use Almgren's frequency function argument. For the full proof, check out Garofalo-Lin's 1987 paper *Unique Continuation for Elliptic Operators: A Geometric-Variational Approach*.

2.2 Carleman Estimates

Carleman estimates, essentially, are a priori estimates in weighted L^2 -norms; it plays a crucial role in proving the uniqueness results in various settings of PDE problems, including our main topic of discussion, backward uniqueness of parabolic equations. Here I would like to **introduce the Carleman estimates and its implications on unique continuation with basic examples**.

We derive Carleman estimates for $\bar{\partial}$ and Δ equations on $\mathbb{R}^2$. Consider the following $\bar{\partial}$ problem on $\mathbb{R}^2$:

$$\begin{cases} \partial_t u + i\partial_x u = a(t, x)u \\ u(t, x) = 0 \text{ for } t < 0 \end{cases} \quad (8)$$

where $a \in L^\infty(\mathbb{R}^2)$. We would like to show that

Theorem 2.4. *The unique solution $u \in C^1(\mathbb{R}_{t,x}^2; \mathbb{C})$ to (8) is $u \equiv 0$.*

To prove this, we need the following estimate:

Lemma 2.5 (Carleman estimates for the $\bar{\partial}$ -equation). *For $\alpha \in (0, \frac{1}{4}]$, $(t, x) \in \mathbb{R}^2$, define $\phi(t, x) = t + \alpha x^2 - \frac{t^2}{2}$. Then, for all $w \in C_c^1(\mathbb{R}^2; \mathbb{C})$, $\lambda \geq 0$, we have*

$$\sqrt{2} \|e^{-\lambda\phi} (\partial_t + i\partial_x) w\|_{L^2(\mathbb{R}^2)} \geq \sqrt{\lambda} \|e^{-\lambda\phi} w\|_{L^2(\mathbb{R}^2)}, \quad (9)$$

$$\sqrt{2} \|e^{-\lambda\phi} (\partial_t - i\partial_x) w\|_{L^2(\mathbb{R}^2)} \geq \sqrt{\lambda} \|e^{-\lambda\phi} w\|_{L^2(\mathbb{R}^2)}. \quad (10)$$

Proof. We only prove (9): (10) can be proven with $\bar{w}$. Defining $v = e^{-\lambda\phi} w$, it amounts to proving that there exists $C > 0$ such that for all $v \in C_c^1(\mathbb{R}^2; \mathbb{C})$ and all $\lambda \geq 0$,

$$C \|e^{-\lambda\phi} (\partial_t + i\partial_x) (e^{\lambda\phi} v)\|_{L^2(\mathbb{R}^2)} = C \|(\partial_t + i\lambda\phi'_x + i\partial_x + \lambda\phi'_t) v\|_{L^2(\mathbb{R}^2)} \geq \sqrt{\lambda} \|v\|_{L^2(\mathbb{R}^2)}.$$

Note that

$$\begin{aligned} \|(\partial_t + i\lambda\phi'_x + i\partial_x + \lambda\phi'_t) v\|_{L^2(\mathbb{R}^2)}^2 &= \|(\partial_t + i\lambda\phi'_x) v\|^2 + \|(i\partial_x + \lambda\phi'_t) v\|^2 \\ &\quad + 2\operatorname{Re}\langle (\partial_t + i\lambda\phi'_x) v, (i\partial_x + \lambda\phi'_t) v \rangle. \end{aligned}$$

We have

$$\begin{aligned} 2\operatorname{Re}\langle (\partial_t + i\lambda\phi'_x)v, (i\partial_x + \lambda\phi'_t)v \rangle &= \langle -(\partial_t + i\lambda\phi'_x)(i\partial_x + \lambda\phi'_t)v + (i\partial_x + \lambda\phi'_t)(\partial_t + i\lambda\phi'_x)v, v \rangle \\ &= \langle \lambda(1 - 2\alpha)v, v \rangle. \end{aligned}$$

Therefore, for $0 < \alpha \leq \frac{1}{4}$,

$$\|(\partial_t + i\lambda\phi'_x + i\partial_x + \lambda\phi'_t)v\|^2 \geq \frac{\lambda}{2} \|v\|^2.$$

□

Proof of Theorem 2.4. Suppose $u \in C^1(\mathbb{R}_{t,x}^2)$ solves the equation (8). Let χ be a smooth cutoff function such that

$$\chi = 1 \text{ on } t^2 + x^2 \leq 1 \text{ and is supported in } t^2 + x^2 \leq \beta^2, \beta > 1 \text{ to be chosen later.}$$

Let $w = \chi u \in C_c^1(\mathbb{R}^2)$. Since $(\partial_t + i\partial_x)u = au$, by Lemma 2.5, we get

$$\begin{aligned} \sqrt{\lambda} \|e^{-\lambda\phi}\chi u\| &\leq C \|e^{-\lambda\phi}[\partial_t + i\partial_x, \chi]u\| + C \|e^{-\lambda\phi}\chi au\| \\ &\leq C \|e^{-\lambda\phi}(\chi'_t + i\chi'_x)u\| + C \|a\|_{L^\infty} \|e^{-\lambda\phi}u\| \\ &\leq C \|e^{-\lambda\phi}|\nabla\chi|u\|_{L^2} + C \|a\|_{L^\infty} \|e^{-\lambda\phi}u\|_{L^2} \end{aligned}$$

where $[\cdot, \cdot]$ denotes the commutator and

$$[\partial_t + i\partial_x, \chi]u = (\partial_t + i\partial_x)(\chi u) - \chi(\partial_t + i\partial_x)u = (\chi'_t + i\chi'_x)u.$$

Note that

$$\operatorname{supp} (\chi'_t + i\chi'_x)u \subset \{1 \leq t^2 + x^2 \leq \beta^2, t \geq 0\} =: K.$$

As a result, on the support of $(\chi'_t + i\chi'_x)u$, we have

$$\begin{aligned} \phi &= t - \frac{t^2}{2} + \alpha x^2 \geq t - t^2\left(\frac{1}{2} + \alpha\right) + \alpha \\ &\geq t - t\beta\left(\frac{1}{2}\alpha\right) + \alpha \geq \alpha > 0 \end{aligned}$$

if we choose β such that

$$1 < \beta \leq \frac{1}{\frac{1}{2} + \alpha}.$$

As a result, we have

$$\left(\lambda^{1/2} - C\|a\|_{L^\infty}\right) \|e^{-\lambda\phi}\chi u\|_{L^2(\mathbb{R}^2)} \leq C\|\nabla\chi\|_{L^\infty(\mathbb{R}^2)} e^{-\lambda\alpha} \|u\|_{L^2(K)}.$$

So when we take $\lambda \geq 4C^2\|a\|_{L^\infty}^2$ sufficiently large, we should have

$$\sqrt{\lambda} \|e^{-\lambda\phi}\chi u\|_{L^2(\mathbb{R}^2)} \leq 2C\|\nabla\chi\|_{L^\infty(\mathbb{R}^2)} e^{-\lambda\alpha} \|u\|_{L^2(K)}.$$

On the other hand, on $t^2 + x^2 \leq \alpha^4$, we have

$$\phi(t, x) \leq t + x^2 - \frac{x^2}{2} \leq \alpha^2 + \alpha^4,$$

and this implies, for $\lambda \geq 4C^2 \|a\|_{L^\infty}^2$, we have

$$\begin{aligned} \sqrt{\lambda} e^{-\lambda(\alpha^4 + \alpha^2)} \|u\|_{L^2(t^2 + x^2 \leq \alpha^4)} &\leq \sqrt{\lambda} \|e^{-\lambda\phi} u\|_{L^2(t^2 + x^2 \leq \alpha^4)} \\ &\leq \sqrt{\lambda} \|e^{-\lambda\phi} \chi u\|_{L^2} \\ &\leq 2C \|\nabla \chi\|_{L^\infty} e^{-\lambda\alpha} \|u\|_{L^2(K)} \end{aligned}$$

entailing

$$\sqrt{\lambda} \|u\|_{L^2(t^2 + x^2 \leq \alpha^4)} \leq 2C \|\nabla \chi\|_{L^\infty} e^{-\lambda\alpha(1 - (\alpha + \alpha^3))}.$$

As $\alpha \in (0, 1/4]$, $\alpha^3 + \alpha < 1$ and consequently, taking $\lambda \rightarrow \infty$, we see that u vanishes on $t^2 + x^2 \leq \alpha^4$. Since the equation (8) is invariant with respect to translation in x , we obtain that u vanishes on $t \leq \alpha^2$, and by a connected set argument, we can conclude that u vanishes on $\mathbb{R}^2$. $\square$

Remark 2.6. From the proof above, we can notice that a careful choice of suitable ϕ is necessary to derive the estimate and consequently the corresponding unique continuation result.

Remark 2.7. The name of the estimate (obviously) originates from the Swedish mathematician Torsten Carleman (1892-1949) who worked in analysis and invented this method in 1939. Nevertheless, there is another ‘‘Carleman inequality’’ for a product-sum of positive real numbers, whose name also comes from the same mathematician. To distinguish, we use the term ‘‘Carleman estimate’’ instead of ‘‘inequality’’.

For the Laplace operator we have the following Carleman estimate:

Lemma 2.8 (Carleman estimate for Δ). *Let α, ϕ as in Lemma 2.5. Then, for all $w \in C_c^2(\mathbb{R}^2)$, $\lambda \geq 0$, we have*

$$4\sqrt{2} \|e^{-\lambda\phi} \Delta w\|_{L^2(\mathbb{R}^2)} \geq \lambda \|e^{-\lambda\phi} w\|_{L^2(\mathbb{R}^2)} + \sqrt{\lambda} \|e^{-\lambda\phi} \nabla w\|_{L^2(\mathbb{R}^2)}. \quad (11)$$

Proof. We have

$$\begin{aligned} e^{-\lambda\phi} \Delta e^{\lambda\phi} &= e^{-\lambda\phi} (\partial_t + i\partial_x) e^{\lambda\phi} e^{-\lambda\phi} (\partial_t - i\partial_x) e^{\lambda\phi} \\ &= e^{-\lambda\phi} (\partial_t - i\partial_x) e^{\lambda\phi} e^{-\lambda\phi} (\partial_t + i\partial_x) e^{\lambda\phi}, \end{aligned}$$

and consequently for $v \in C_c^2(\mathbb{R}^2)$, $\lambda \geq 0$, we obtain from Lemma 2.5,

$$\begin{aligned} 4\sqrt{2} \|e^{-\lambda\phi} \Delta e^{\lambda\phi} v\|_{L^2(\mathbb{R}^2)} &\geq 2\sqrt{\lambda} \|e^{-\lambda\phi} (\partial_t - i\partial_x) e^{\lambda\phi} v\|_{L^2} + 2\sqrt{\lambda} \|e^{-\lambda\phi} (\partial_t + i\partial_x) e^{\lambda\phi} v\|_{L^2} \\ &\geq \sqrt{\lambda} \|e^{-\lambda\phi} \partial_t e^{\lambda\phi} v\|_{L^2} + \sqrt{\lambda} \|e^{-\lambda\phi} \partial_x e^{\lambda\phi} v\|_{L^2} + 2\sqrt{\lambda} \sqrt{\lambda/2} \|v\|_{L^2} \\ &\geq \sqrt{\lambda} \|e^{-\lambda\phi} \nabla e^{\lambda\phi} v\|_{L^2} + \lambda \|v\|_{L^2}, \end{aligned}$$

proving the lemma with $v = we^{-\lambda\phi}$. $\square$

Repeating a similar argument as the $\bar{\partial}$ equation case, we can use [Lemma 2.8](#) to prove the uniqueness of the following equation:

$$\begin{cases} \partial_t^2 u + \partial_x^2 u = a_0 u + b_1 \partial_t u + c_1 \partial_x u \\ u(t, x) = 0 \text{ for } t < 0 \end{cases} \quad (12)$$

where $a_0, b_1, c_1 \in L^\infty(\mathbb{R}^2)$.

Theorem 2.9. *The unique solution $u \in C^2(\mathbb{R}_{t,x}^2; \mathbb{C})$ to (12) is $u \equiv 0$.*

2.3 Backward Uniqueness of Parabolic Equations

Now we give a sketch of proof of [Theorem 2.1](#). As previously mentioned, we will need some Carleman-type estimates to proceed.

Lemma 2.10 (Carleman estimate for backward uniqueness, 1). *Let $\sigma(t) = te^{-\frac{t}{8}}$, $\sigma_a(t) = \sigma(t+a)$. Then $\exists N = N(n)$ such that*

$$\begin{aligned} & \left\| \sigma_a^{-\alpha-\frac{1}{2}} e^{-\frac{|x-y|^2}{8(t+a)}} u \right\|_{L^2(\mathbb{R}^n \times (0,1))} + \left\| \sigma_a^{-\alpha} e^{-\frac{|x-y|^2}{8(t+a)}} \nabla u \right\|_{L^2(\mathbb{R}^n \times (0,1))} \\ & \leq N \left\| \sigma_a^{-\alpha} e^{-\frac{|x-y|^2}{8(t+a)}} (\Delta u + \partial_t u) \right\|_{L^2(\mathbb{R}^n \times (0,1))} \end{aligned} \quad (13)$$

holds for any $\alpha \geq 0$, $y \in \mathbb{R}^n$, $0 < a < 1$, $u \in C_0^\infty(\mathbb{R}^n \times [0,1))$ with $u(\cdot, 0) \equiv 0$.

Lemma 2.11 (Carleman estimate for backward uniqueness, 2). $\exists$ constant $\alpha_0 = \alpha_0(R, n)$ such that the estimate

$$\begin{aligned} & \left\| e^{\alpha(T-t)(|x|-R)+|x|^2} u \right\|_{L^2(Q_{R,T})} + \left\| e^{\alpha(T-t)(|x|-R)+|x|^2} \nabla u \right\|_{L^2(Q_{R,T})} \\ & \leq \left\| e^{\alpha(T-t)(|x|-R)+|x|^2} (\Delta u + \partial_t u) \right\|_{L^2(Q_{R,T})} + \left\| e^{|x|^2} \nabla u(\cdot, T) \right\|_{L^2(\mathbb{R}^n - \bar{B}_R)} \end{aligned} \quad (14)$$

holds for $\alpha \geq \alpha_0$ and $u \in C_0^\infty(Q_{R,T})$ with $u(\cdot, 0) \equiv 0$.

Proof of Lemma 2.10, 2.11. See Escauriaza-Seregin-Šverák's original paper. A similar derivation of Carleman estimate is also presented in Wang's paper. $\square$

Estimates (13) and (14) are essential for proving [Theorem 2.1](#) as they lead up to the following key lemma (with the help of standard parabolic estimates):

Lemma 2.12. *Assume that u satisfies in $(\mathbb{R}^n \setminus \bar{B}_R) \times [0, 1]$*

$$|\Delta u + \partial_t u| \leq \epsilon (|u| + |\nabla u|), \quad |u(x, t)| \leq e^{\epsilon|x|^2}, \quad (15)$$

and $u(x, 0) \equiv 0$ in $\mathbb{R}^n \setminus \bar{B}_R$ for some $R \geq 1$. Then, there is $\epsilon(n) > 0$ such that u is identically zero in $(\mathbb{R}^n \setminus \bar{B}_R) \times [0, \epsilon]$ when $\epsilon \leq \epsilon(n)$.

Proof of Theorem 2.1. If u satisfies the assumptions in [Theorem 2.1](#), then for some $\delta > 0$, the function $u_0 = u(\delta x, \delta^2 t)/M$ also satisfies (15) in $Q_{R,T}$ with arbitrarily small ϵ with new $R, T \geq 1$.

Then by [Lemma 2.12](#), we would have $u_0 \equiv 0$ in $(\mathbb{R}^n \setminus \bar{B}_R) \times [0, \epsilon]$. Since $u_0(x, t + \epsilon)$ satisfies (15) in $(\mathbb{R}^n \setminus \bar{B}_R) \times [0, T - \epsilon]$, we get $u_0 \equiv 0$ in $(\mathbb{R}^n \setminus \bar{B}_R) \times [0, 2\epsilon]$ and so on. $\square$

Proof of Lemma 2.12. We will need two fundamental estimates in parabolic PDE theory (ref. Ladyzhenskaya):

Lemma 2.13 (Gradient estimate for heat equation). *Let $P_r = B_r \times [-r^2, 0]$. There exists $C = C(n) < \infty$ such that, for any $u \in C^\infty(P_r)$, we have*

$$\sup_{(x,t) \in P_r} d_{x,t} |\nabla u(x,t)| \leq C(n) \sup_{(x,t) \in P_r} (|u(x,t)| + d_{x,t}^2 |(\partial_t + \Delta) u(x,t)|)$$

where $d_{x,t} = \min\{d(x, \partial B_r), |t|\}$.

Lemma 2.14 (Parabolic mean value inequality). *There exists $C = C(n)$ such that, for any $s > 0$ and $u \in C^\infty(B(y, \sqrt{s}) \times [s, 2s])$ satisfying $|(\partial_t + \Delta) u| \leq |\nabla u| + |u|$, we have*

$$|u(y, s)|^2 \leq \frac{C}{s^{n+2}} \int_s^{2s} \int_{B(y, \sqrt{s})} u^2 dx dt.$$

We first show that there exists $\epsilon(n) > 0$ and a constant $N = N(n)$ such that any such function satisfies, in $\mathbb{R}^n \setminus B_{6R} \times [0, \frac{1}{N}]$,

$$|u(y, s)| + |\nabla u(y, s)| \leq N e^{\frac{-|y|^2}{Ns}} \left(1 + \|u\|_{L^\infty(B_{4R} \setminus B_R \times [0, 1])}\right) \quad (16)$$

where $\epsilon \leq \epsilon(n)$.

Assuming that u satisfies (15), $|y| \geq 6R$, $a \in (0, 1)$ and that $r \geq 4|y|$ is a large number, we will apply [Lemma 2.10](#) to $u_r(x, t) = u(x, t)\phi(t)\psi_r(x)$, where $\phi \in C^\infty(\mathbb{R})$ and $\psi_r \in C_0^\infty(\mathbb{R}^n)$ satisfy $\phi = 1$ for $t \leq \frac{1}{2}$, $\phi = 0$ for $t \geq \frac{3}{4}$, $\psi_r = 1$ for $3R \leq |x| \leq 2r$, $\psi_r = 0$ for $|x| \leq 2R$ and $|x| \geq 3r$, $0 \leq \phi, \psi \leq 1$, and taking $\alpha = k \in \mathbb{N}$ in (13). Then we have

$$\begin{aligned} & \left\| \sigma_a^{-k-\frac{1}{2}} e^{\frac{-|x-y|^2}{8(t+a)}} u_r \right\|_2 + \left\| \sigma_a^{-k} e^{\frac{-|x-y|^2}{8(t+a)}} \nabla u_r \right\|_2 \lesssim \left\| \sigma_a^{-k} e^{\frac{-|x-y|^2}{8(t+a)}} (\Delta + \partial_t) u_r \right\|_2 \\ & \lesssim \left\| \sigma_a^{-k} e^{\frac{-|x-y|^2}{8(t+a)}} u \right\|_{L^2(\mathbb{R}^n \setminus B_R \times [\frac{1}{2}, \frac{3}{4}])} + \left\| \sigma_a^{-k} e^{\frac{-|x-y|^2}{8(t+a)}} (|u| + |\nabla u|) \right\|_{L^2(B_{3R} \setminus B_{2R} \times [0, \frac{3}{4}])} \\ & \quad + \left\| \sigma_a^{-k} e^{\frac{-|x-y|^2}{8(t+a)}} (|u| + |\nabla u|) \right\|_{L^2(B_{3r} \setminus B_{2r} \times [0, \frac{3}{4}])} \end{aligned} \quad (17)$$

By the growth assumption of u given in [Theorem 2.1](#) plus [Lemma 2.13](#), we can obtain the bound for the last term on the RHS of (17):

$$\left\| \sigma_a^{-k} e^{\frac{-|x-y|^2}{8(t+a)}} (|u| + |\nabla u|) \right\|_{L^2(B_{3r} \setminus B_{2r} \times [0, \frac{3}{4}])} \lesssim a^{-k} e^{-r^2} \|u\|_{L^\infty(B_{4r} \setminus B_r \times [0, 1])} \rightarrow 0 \text{ as } r \rightarrow \infty \text{ when } \epsilon < \frac{1}{16}.$$

Letting $r \rightarrow \infty$ and $a \rightarrow 0$ on the RHS of (17), following from (15), there exists $N(n) = N$ such

that, for $k \geq 0$,

$$\begin{aligned} & \left\| t^{-k-\frac{1}{2}} e^{-\frac{|x-y|^2}{8t}} u \right\|_{L^2(\mathbb{R}^n \setminus B_{2R} \times [0, \frac{1}{2}])} + \left\| t^{-k} e^{-\frac{|x-y|^2}{8t}} \nabla u \right\|_{L^2(\mathbb{R}^n \setminus B_{2R} \times [0, \frac{1}{2}])} \\ & \leq N^k \left\| e^{-\frac{|x-y|^2}{8t}} u \right\|_{L^2(\mathbb{R}^n \setminus B_R \times [\frac{1}{2}, \frac{3}{4}])} + N^k \left\| t^{-k} e^{-\frac{|x-y|^2}{8t}} (|u| + |\nabla u|) \right\|_{L^2(B_{3R} \setminus B_{2R} \times [0, \frac{3}{4}])}. \end{aligned} \quad (18)$$

From the growth condition (15), we have

$$\left\| e^{-|x-y|^2/8t} u \right\|_{L^2(\mathbb{R}^n \setminus B_R \times [\frac{1}{2}, \frac{3}{4}])} \lesssim \left\| e^{\epsilon|x|^2 - \frac{|x-y|^2}{6}} \right\|_{L^2(\mathbb{R}^n)} \lesssim e^{|y|^2}. \quad (19)$$

On the other hand, using Stirling's formula and [Lemma 2.13](#), we can deduce that

$$\left\| t^{-k} e^{-\frac{|x-y|^2}{8t}} (|u| + |\nabla u|) \right\|_{L^2(B_{3R} \setminus B_{2R} \times [0, \frac{3}{4}])} \leq N^k k! |y|^{-2k} \|u\|_{L^\infty(B_{4R} \setminus B_R \times [0, 1])}. \quad (20)$$

Again making use of the growth condition (15) and estimates above, we have an improved estimate of (18): there exists $N = N(n)$ such that, for $|y| \geq 6R$ and $k \geq 0$,

$$\left\| t^{-k} e^{-\frac{|x-y|^2}{8t}} u \right\|_{L^2(\mathbb{R}^n \setminus B_{2R} \times [0, \frac{1}{2}])} \leq N^k (k! |y|^{-2k} \|u\|_{L^\infty(B_{4R} \setminus B_R \times [0, 1])} + e^{|y|^2}),$$

which yields

$$\left\| e^{\frac{|y|^2}{4Nt}} e^{-\frac{|x-y|^2}{8(t+a)}} \right\|_{L^2(\mathbb{R}^n \setminus B_{2R} \times [0, \frac{1}{8N}])} \lesssim 1 + \|u\|_{L^\infty(B_{4R} \setminus B_R \times [0, 1])}.$$

Applying [Lemma 2.13](#) and [Lemma 2.14](#), we obtain (16).

Now we apply [Lemma 2.11](#) to $u_{a,r} = u \cdot \psi_{a,r}$ where $a \in (0, 1)$, $r > 0$ large,

$$\psi_{a,r}(x) = \begin{cases} = 0, & |x| \leq (1+a)R \text{ or } |x| \geq 2r \\ = 1, & (1+2a)R \leq |x| \leq r \end{cases} \in C_c^\infty(\mathbb{R}^n)$$

and choose $T = 4\epsilon$ with $0 < \epsilon \leq \frac{1}{10N}$. Then we have

$$\begin{aligned} & \left\| e^{\alpha(4\epsilon-t)(|x|-R)} + |x|^2 u_{a,r} \right\|_{L^2(Q_{R,4\epsilon})} + \left\| e^{\alpha(4\epsilon-t)(|x|-R)} + |x|^2 \nabla u_{a,r} \right\|_{L^2(Q_{R,4\epsilon})} \\ & \leq \left\| e^{\alpha(4\epsilon-t)(|x|-R)} + |x|^2 (\Delta + \partial_t)(u_{a,r}) \right\|_{L^2(Q_{R,4\epsilon})} \\ & \quad + \left\| e^{|x|^2} \nabla u_{a,r}(\cdot, 4\epsilon) \right\|_{L^2(\mathbb{R}^n)}. \end{aligned} \quad (A)$$

As before,

$$|(\Delta + \partial_t)(u_{a,r})| \lesssim \epsilon (|u_{a,r}| + |\nabla u_{a,r}|) + |u| (|(\Delta \psi_{a,r}| + |\nabla \psi_{a,r}|) + 2|\nabla \psi_{a,r}| |\nabla u|), \quad (21)$$

and the first term on the right-hand side of (21) can be hidden in the left-hand side of (A) when ϵ is sufficiently small. These imply that the following inequality holds with a constant depending on $a > 0$:

$$\begin{aligned}
e^{10\alpha\varepsilon aR} \|u\|_{L^2(B_r \setminus B_{(1+10a)R} \times [0, \varepsilon])} &\lesssim e^{8\alpha\varepsilon r + 4r^2} \| |u| + |\nabla u| \|_{L^2(B_{2r} \setminus B_r \times [0, 4\varepsilon])} \\
&+ e^{8\alpha\varepsilon aR} \| |u| + |\nabla u| \|_{L^2(B_{(1+2a)R} \setminus B_{(1+a)R} \times [0, 4\varepsilon])} \\
&+ \|e^{|x|^2} u(\cdot, 4\varepsilon)\|_{L^2(B_{2r} \setminus B_{(1+a)R})} \\
&+ \|e^{|x|^2} \nabla u(\cdot, 4\varepsilon)\|_{L^2(B_{2r} \setminus B_{(1+a)R})}.
\end{aligned}$$

This inequality and (2.8) imply that for some constant depending on a and ε ,

$$\|u\|_{L^2(B_r \setminus B_{(1+10a)R} \times [0, \varepsilon])} \lesssim e^{\alpha r - r^2} + e^{-2\alpha\varepsilon aR}.$$

Then, letting first $r \rightarrow +\infty$, then $\alpha \rightarrow +\infty$ and finally $a \rightarrow 0^+$ in the last inequality, gives $u \equiv 0$ in $\mathbb{R}^n \setminus B_R \times [0, \varepsilon]$. □

3 Back to Geometry (week 2)

3.1 Notions/Definitions

Definition 3.1 (Self-shrinker). *A hypersurface Σ in $\mathbb{R}^{n+1}$ is said to be a **self-shrinker** if it satisfies*

$$H = \operatorname{div}(\mathbf{n}) = \frac{1}{2} \langle x, \mathbf{n} \rangle$$

where $\mathbf{n}$ is the outward unit normal of Σ , x is the position vector, $\langle \cdot, \cdot \rangle$ is the Euclidean inner product.

- Since Σ is a self-shrinker, we know that $\sqrt{-t}\Sigma$, $t \in (0, 1]$ satisfies the (forward) MCF, which implies that $\Sigma_t = \sqrt{t}\Sigma$ satisfies the backward MCF equation for $t \in (0, 1]$:

$$-\partial_t x = H\mathbf{n} \tag{22}$$

- Studying geometric structures of self-shrinkers is of great interest because shrinkers belong a type of limit solutions to the MCF (more specific, it is a Type-I singularity model of MCF (Huisken)).

Definition 3.2 (Regular cone). *We say that $C \subset \mathbb{R}^{n+1}$ is a **regular cone** with vertex at O , if*

$$C = \{l\Gamma : 0 \leq l < \infty\}$$

where Γ is a smooth, connected, closed embedded submanifold of S_1 with codimension 1.

3.2 Geometric properties of asymptotically conical self-shrinker

We first discuss some useful geometric properties of the asymptotically conical structure of the shrinkers Σ_t .

We will often use a modified version of the below geometric lemma:

Lemma 3.3 (Small curvature implies graphical, SCIG). *Let $\Sigma \subset \mathbb{R}^n$ be an immersed hypersurface with*

$$16s^2 \cdot \sup_{\Sigma} |A_{\Sigma}|^2 \leq 1.$$

If $x \in \Sigma$ and $d_{\Sigma}(x, \partial\Sigma) \geq 2s$, then there exists a function u defined on $B_s^T(x) = B_s^{\Sigma}(x) \cap T_x\Sigma$ such that $B_s^{\Sigma}(x) \subset \text{Graph}(u) \subset B_{2s}^{\Sigma}(x)$.

Moreover,

1. $|\nabla u| \leq 1$ and $\sqrt{2}s |\text{Hess } u| \leq 1$;
2. the connected component of $B_s(x) \cap \Sigma$ containing x is contained in $B_{2s}(x)$.

Proof. We only present an outline of the proof. The argument used here will also be repeated (without modification) in later discussions.

1. Using the Gauss map $N : B \rightarrow \mathbb{S}^{n-1}$, define the angular distance on B , where B is some small ball $B(x, 2s)$ contained in Σ . Then along a geodesic connecting x and $y \in B$, we find that the angular distance between x and y is less than $\frac{1}{2}$ by the given bound on the second fundamental form A of Σ ($|dN| \leq |A|$).
2. And since $\frac{1}{2} < \frac{\pi}{4} (< \frac{\pi}{2})$, we can write B as a graph of a function u over some $B_0 \subset T_x\Sigma$ containing 0. (Imagine the case where Σ = sphere) In fact, (locally) u is the inverse of nearest point projection from the tangent space $T_x\Sigma$ to Σ .
3. The normal vector $\mathbf{n}$ over Σ is written as

$$\mathbf{n} = \frac{(-\nabla u, 1)}{\sqrt{1 + |\nabla u|^2}}$$

where $\mathbf{n}(x) = (0, \dots, 0, 1)$. Using this representation of $\mathbf{n}$ we can obtain the desired bounds as follows:

$$\begin{aligned} 1 + |\nabla u|^2 &= \langle \mathbf{n}(x), \mathbf{n}(y) \rangle^{-2} = |\mathbf{n}(x)|^{-2} |\mathbf{n}(y)|^{-2} \cos(\theta)^{-2} \\ &= \cos(\theta)^{-2} \leq 2 \\ \implies |\nabla u|^2 &\leq 1. \end{aligned}$$

Hessian bound is given by the below elementary inequality (it is clear once you compute the induced metric of surface parametrized as a graph of u ; for simplicity, consider the 2-dimensional case):

$$|\text{Hess } u|^2 \leq |A|^2 \left(1 + |\nabla u|^2\right)^3 \leq \frac{1}{2s^2}.$$

This completes the proof of (1).

4. For (2), we need to show $\partial B_{2s}(x)$ lies outside of $B_s^{\Sigma}(x)$. This is done by taking a geodesic and again making use of the fact that $|dN| \leq |A|$.

□

Lemma 3.4 (Bounds on $|\nabla^i A|$). *There exists $C_1 > 0$ and $R_1 \geq R_0$ such that, for all $x \in \Sigma_t \setminus B_{R_1}$, $t \in (0, 1]$, $0 \leq i \leq 2$,*

$$|\nabla^i A(x)| \leq C_1 |x|^{-1-i} \quad (23)$$

where $A = A_{\Sigma_t}$ is the second fundamental form on Σ_t .

Proof. We make use of the C_{loc}^∞ -convergence of $\lambda\Sigma$ to C to obtain control over the geometry of Σ and Σ_t as follows:

- Since Γ is a smooth closed embedded submanifold of S_1 , the second fundamental form and all its covariant derivatives of C are bounded inside the annulus $C \cap (\bar{B}_2 \setminus B_{\frac{1}{2}})$.
- Furthermore, since $\lambda\Sigma$ converges to the regular cone C in the C_{loc}^∞ sense as $\lambda \rightarrow 0^+$, there exists $\delta_1, \lambda_1 > 0$ such that for $0 < \lambda < \delta_1$, $0 \leq i \leq 2$,

$$|\nabla^i A| \leq \lambda_1 \quad (A = A_\Sigma)$$

on $\lambda\Sigma \cap (\bar{B}_2 \cap B_{\frac{1}{2}})$. Therefore, for $x \in \Sigma$ with $|x| \geq R_1 = 2 \max\{R_0, \delta^{-1}\}$, we choose $\lambda = \frac{1}{|x|}$ to see that, for $i = 0, 1, 2$,

$$|\nabla^i A(x)| = \lambda^{i+1} |\nabla^i A(\lambda x)| \leq \lambda_1 |x|^{-1-i}. \quad (24)$$

- Since $\Sigma_t = \sqrt{t}\Sigma$, we conclude that, for $x \in \Sigma_t \setminus B_{R_1}$ and $i = 0, 1, 2$,

$$|\nabla^i A(x)| = \sqrt{t}^{-1-i} \left| \nabla^i A \left(\frac{x}{\sqrt{t}} \right) \right| \leq \lambda_1 |x|^{-1-i}. \quad (25)$$

Then choose $C_1 = \lambda_1$.

□

Now we use the idea from [Lemma 3.3](#) to prove the following important result:

Lemma 3.5 (Graphical representation of Σ_t over C). *There exists $R_2 > R_1$ and compact sets $K_t \subset \Sigma_t$, $0 < t \leq 1$, such that $K_t \subset \Sigma_t \cap B_{2R_2}$ and $\Sigma_t \setminus K_t$ is given by the graph of a smooth function $U(\cdot, t) : C \setminus \bar{B}_{R_2} \rightarrow \mathbb{R}$. Moreover, there exists $0 < \epsilon_0 < 1$ and $C_2 > 0$ such that, for $z_0 \in C \setminus B_{2R_2}$ and $t \in (0, 1]$, the component of $\Sigma_t \cap B_{\epsilon_0|z_0|}(z_0)$ containing $z_0 + U(z_0, t)$ can be written as the graph of a smooth function $u(\cdot, t)$ over the tangent hyperplane $T_{z_0}C$ of C at z_0 satisfying that, for $i = 0, 1, 2$,*

$$|D^{i+1}u| \leq C_2 |z_0|^{-i} \quad \text{and} \quad |D^i \partial_t u| \leq C_2 |z_0|^{-1-i} \quad (26)$$

where D, D^2 are the Euclidean gradient and Hessian on $\mathbb{R}^n \cong T_{z_0}C$ and ∂_t denotes the partial derivative with respect to t fixing points in $T_{z_0}C$.

Proof. e

- Given a small $\delta > 0$, by the C_{loc}^∞ convergence $\lambda\Sigma \rightarrow C$, we know that there exists $t_0 = t_0(\delta) \in (0, R_1^{-2})$ such that, if $0 < t \leq t_0$, then $\Sigma_t \cap B_3 \cap B_{\frac{1}{3}}$ can be written as the graph of a function $V(\cdot, t) : \Omega_t \rightarrow \mathbb{R}$ satisfying $\|V(\cdot, t)\|_{C^1} \leq \delta$. Here, the domain Ω_t of $V(\cdot, t)$ satisfies

$$(B_2 \setminus B_{\frac{1}{2}}) \cap C \subset \Omega_t \subset (B_4 \setminus B_{\frac{1}{4}}) \cap C \quad \text{and} \quad \Omega_t \rightarrow (B_3 \setminus B_{\frac{1}{3}}) \cap C.$$

- Then we can choose δ sufficiently small, depending on C inside the annulus $\bar{B}_4 \setminus B_{\frac{1}{4}}$ such that, for $z \in (B_2 \setminus B_{\frac{1}{2}}) \cap C$, $x_t = z + V(z, t)\mathbf{n}(z) \in \Sigma_t$ and $0 < t \leq t_0$, the distance $\text{dist}(x_t, C)$, which is achieved uniquely at z and is equal to $V(z, t)$, is less than $\frac{1}{100}$ and $\langle \mathbf{n}(x_t), \mathbf{n}(z) \rangle > \frac{99}{100}$, which is possible given the upper bound on the C^1 -norm of V .
- Now suppose $|x|^2 > t_0^{-1}$, $x \in \Sigma_t$, $t_0 \in (0, 1]$. Then $\frac{x}{|x|} \in B_2 \setminus B_{\frac{1}{2}}$ and the homogeneity of C yields

$$\text{dist}(x, C) < \frac{|x|}{100} :$$

We can choose sufficiently small $t_0 \in (0, 1)$ such that

$$\frac{1}{t_0} < |x|^2 < \frac{4}{t_0}.$$

$\implies \sqrt{t_0}x \in \Sigma_{t_0 t} = \sqrt{t_0}t\Sigma$ and $t_0 t < 1$ with $1 < |\sqrt{t_0}x| \leq 2$. Therefore, $\text{dist}(\sqrt{t_0}x, C) < \frac{1}{100}$. Then we use the **homogeneity of the cone** C to obtain:

$$\text{dist}(x, C) < \frac{1}{100\sqrt{t_0}} \leq \frac{|x|}{100} \text{ with } \langle \mathbf{n}(x), \mathbf{n}(z) \rangle > \frac{99}{100} \quad (27)$$

where z is the unique point on C with $\text{dist}(x, C) = \text{dist}(z, C)$.

- By the above discussion, we find that the nearest point projection $\Pi_t : \Sigma_t \setminus B_{\frac{2}{\sqrt{t_0}}} \rightarrow C$ is well-defined for each $t \in (0, 1]$ and

$$\frac{|x|}{2} < |\Pi_t(x)| < 2|x|$$

for $x \in \Sigma_t \setminus B_{\frac{2}{\sqrt{t_0}}}$. Moreover, the map Π_t is injective and the image of Π_t containing $C \setminus \bar{B}_{\frac{4}{\sqrt{t_0}}}$. Then we take $R_2 = \frac{4}{\sqrt{t_0}}$, $K_t = \Sigma_t \setminus \Pi_t^{-1}(C \setminus \bar{B}_{\frac{4}{\sqrt{t_0}}})$ and $U(\cdot, t) = \Pi_t$.

- Now we utilize the arguments in the proof [Lemma 3.3](#), SCIG, to show our conclusion. By the above discussion, we know that given $\delta > 0$, there exists $r = r(\delta) > 0$ such that for $z \in C \setminus B_r$ and $t \in (0, 1]$,

$$|\Pi_t^{-1}(z_0) - z_0| < \delta |z_0|.$$

Recall that in [Lemma 3.4](#) we have shown

$$\frac{1}{C_1} |A(x)| |x| \leq 1.$$

By modifying the argument in SCIG, we can conclude: there exists $\delta_1 \in (0, 1)$, $r_1, \lambda_1 > 0$, depending only on C_1 , such that for $z_0 \in C \setminus B_{r_1}$ and $t \in (0, 1]$, the component of $\Sigma_t \cap B_{\delta_1 |z_0|}(z_0)$ containing $\Pi_t^{-1}(z_0)$ can be written as the graph of a smooth function $u(\cdot, t)$ over $T_{z_0}C$ and $u(\cdot, t)$ satisfies

$$|Du(\cdot, t)| \leq \lambda_1 \text{ and } |D^2u(\cdot, t)| \leq \lambda_1 |z_0|^{-1}. \quad (28)$$

- Using the Simon's identity and [Lemma 3.4](#) again, we can obtain

$$|D^3 u(\cdot, t)| \leq \lambda_2 |z_0|^{-2} \quad (29)$$

$$|D^4 u(\cdot, t)| \leq \lambda_2 |z_0|^{-3} \quad (30)$$

where $\lambda_2 = \lambda_2(C_1, \lambda_1)$.

- Below we prove the bounds on $\partial_t u$. Note that, since Σ_t varies smoothly with respect to t , by the construction of the function u , we know that u varies smoothly with respect to t . We parametrize $T_{z_0} C$ by $F : \mathbb{R}^n \rightarrow T_{z_0} C$, $p \mapsto z_0 + \sum_i p^i e_i$ and identify $u(p, t)$ with $u(F(p), t)$.
- We utilize the backward MCF equation to obtain the desired bounds. Since $\Sigma_t = \sqrt{t}\Sigma$ flows by the mean curvature backwards, we have

$$-\partial_t u = \sqrt{1 + |Du|^2} \operatorname{div} \left(\frac{Du}{\sqrt{1 + |Du|^2}} \right) \quad (31)$$

Expanding the divergence term, we see that, by the estimates (26), we have $|\partial_t u| \leq \lambda_3 |z_0|^{-1}$, where $\lambda_3 = \lambda_3(\lambda_1)$. Differentiating the equation (31) back and forth, combined with the estimates (28)-(30), we obtain the bounds on higher derivatives of $\partial_t u$ as well.

□

Now that we have represented the shrinkers Σ and $\tilde{\Sigma}$ as graphs over fixed hyperplanes tangent to C , we want to show that, outside a compact set, we can represent $\tilde{\Sigma}$ as a graph over Σ .

Lemma 3.6 ($\tilde{\Sigma}$ as a graph of Σ outside a compact set). *There exist $R_3 > R_2$ and $C_3 > 0$ such that, outside a compact set, $\tilde{\Sigma}$ is given by the graph of a smooth function $v : \Sigma \setminus \bar{B}_{R_3} \rightarrow \mathbb{R}$ satisfying that for $x \in \Sigma \setminus \bar{B}_{R_3}$ and $i = 0, 1, 2$,*

$$|\nabla_{\Sigma}^i v(x)| \leq C_3 |x|^{-1-i}. \quad (32)$$

Proof. We only give an outline of the proof. The idea is similar to that of [Lemma 3.5](#).

- Applying [Lemma 3.5](#) to $\tilde{\Sigma}$, we can see that $\tilde{\Sigma}$, outside a compact set, can be written as a graph of a smooth function $\tilde{u}$ over the hyperplane $T_{z_0} C$ satisfying derivative bounds (24).
- Since both $u, \tilde{u}$ are defined on the same hyperplane and are constructed with the nearest point projections Π and $\tilde{\Pi}$, we can apply [Lemma 3.4](#) along SCIG to show that in fact, outside a large ball B_{r_5} , the nearest point projection

$$\tilde{\Sigma} \setminus B_{4r_1} \rightarrow \Sigma$$

is well-defined, injective, and its image contains $\Sigma \setminus B_{r_2}$. (Note that r_5 depends on C_2 and $\tilde{C}_2$.) Then $\tilde{\Sigma}$ is given by the graph of a smooth function v over $\Sigma \setminus \bar{B}_{r_5}$.

- Derivative bounds (32) is given by the backwards MCF equation, similar to the proof in [Lemma 3.6](#).

□

3.3 Reformulation of the Problem and Carleman Estimates

Following the estimates given in section 3.2, we can formulate our problem as an elliptic PDE uniqueness problem as below.

Lemma 3.7 (Elliptic PDE reformulation). *There exists $C_4 > 0$ such that at $x \in \Sigma \setminus B_{R_3}$,*

$$L_\Sigma v = \Delta_\Sigma v - \frac{1}{2} \langle x, \nabla_\Sigma v \rangle + \left(|A|^2 + \frac{1}{2} \right) v + Q(x, v, \nabla_\Sigma v, \nabla_\Sigma^2 v) = 0 \quad (33)$$

where the function Q satisfies

$$|Q(x, v, \nabla_\Sigma v, \nabla_\Sigma^2 v)| \leq C_4 |x|^{-2} (|v| + |\nabla_\Sigma v|). \quad (34)$$

Proof. Equation (33) is equivalent from the shrinker equation of $\tilde{\Sigma}$, expressed in terms of v . To write out the equation in terms of v and to obtain the bound on Q , we want to parametrize everything and write them out in the most elementary manner possible.

- Fix $x_0 \in \Sigma \setminus B_{R_3}$. Choose a local parametrization of Σ in a neighborhood of x_0 , $F : \Omega \rightarrow \Sigma$, satisfying $F(0) = x_0$, $\langle \partial_i F(0), \partial_j F(0) \rangle = \delta_{ij}$ and $\partial_{ij}^2 F(0) = a_{ij}(0) \mathbf{n}(x_0)$ with $a_{ij} = A(\partial_i F, \partial_j F)$ and $a_{ij}(0) = 0$ if $i \neq j$. Here Ω is a domain in $\mathbb{R}^n$ containing 0.
- Thus, in a neighborhood of $y_0 = x_0 + v(x_0) \mathbf{n}(x_0)$, there exists a local parametrization of $\tilde{\Sigma}$, such that for $p \in \Omega$,

$$\tilde{F}(p) = F(p) + v(F(p)) \mathbf{n}(F(p)).$$

Then we have

$$\partial_i \tilde{F} = \partial_i F + (\partial_i v) \mathbf{n} + v \partial_i \mathbf{n}$$

and

$$\partial_{ij}^2 \tilde{F} = \partial_{ij}^2 F + (\partial_{ij}^2 v) \mathbf{n} + (\partial_i v) \partial_j \mathbf{n} + (\partial_j v) \partial_i \mathbf{n} + v \partial_{ij}^2 \mathbf{n}.$$

- Denote $g = g_{ij}$ the pullback metric on $\tilde{\Sigma}$ induced by $\tilde{F} = F + v \mathbf{n} \circ F$:

$$g_{ij} = \langle \partial_i F, \partial_j F \rangle = (1 - a_{ii} v)(1 - a_{jj} v) \delta_{ij} + \partial_i v \partial_j v.$$

Then at $p = 0$, we have

$$\det(g) = 1 - 2 \sum_k a_{kk} v + Q_1(p, v, Dv) \quad (35)$$

and

$$g^{ij} \det(g) = \begin{cases} Q_{2ij}(p, v, Dv) \partial_i v & \text{if } i \neq j \\ 1 - 2 \sum_{k \neq i} a_{kk} v + Q_{2ij}(p, v, Dv) & \text{if } i = j. \end{cases} \quad (36)$$

Then, by [Lemma 3.6](#) we have the bounds

$$|Q_1(0, v, Dv)| \leq \lambda_1 |x_0|^{-2} (|v(0)| + |Dv(0)|)$$

and similar ones for $|Q_{2ij}(0, v, Dv)|$.

- Recall the backward shrinker equation

$$\sum_{i,j} g^{ij} \langle \partial_{ij} \tilde{F}, \mathbf{N} \rangle = -\frac{1}{2} \langle \tilde{F}, \mathbf{N} \rangle \quad (37)$$

where

$$\mathbf{N} = - \sum_k \left[\prod_{l \neq k} (1 - a_{kl} v) \right] (\partial_k v) \partial_k F + \left[\prod_k (1 - a_{kk} v) \right] \mathbf{n}$$

is a vector parallel to the unit normal vector $\mathbf{n}(\tilde{F}(0)) \in \tilde{\Sigma}$. Putting all the estimates back into the above shrinker equation, at $p = 0$ we have

$$\sum_k \partial_{kk} v - \frac{1}{2} \langle F, \partial_k F \rangle \partial_k v + \left(|A|^2 + \frac{1}{2} \right) v = \tilde{Q}(0, v, Dv, D^2 v) \quad (38)$$

where $\tilde{Q}$ satisfies

$$|\tilde{Q}(0, v, Dv, D^2 v)| \leq \lambda_2 |x_0|^{-2} (|v(0)| + |Dv(0)|).$$

Since we have chosen a "normal" coordinate + neighborhood at x_0 , we have derived the desired elliptic PDE formulation. □

Now we know that the original problem has been reformulated as a unique continuation problem of the elliptic equation derived in [Lemma 3.7](#). Nevertheless, even in the simplest case $\Sigma = \mathbb{R}^n \times \{0\}$, we cannot utilize the strong unique continuation theorem (by Garofalo-Lin) since the solution cannot be expected to meet the "vanishing of infinite order" condition. Hence, Wang rewrites the elliptic equation (33) as a parabolic equation using the property of shrinker.

Define a function $w : \bigcup_{t \in (0,1]} \Sigma_t \times \{t\} \rightarrow \mathbb{R}$ by

$$w(x, t) = \sqrt{t} v \left(\frac{x}{\sqrt{t}} \right).$$

Then $w(x, t)$ automatically satisfies

$$\frac{dw}{dt} + \Delta_{\Sigma_t} w = \frac{1}{2\sqrt{t}} v \left(\frac{x}{\sqrt{t}} \right) + \sqrt{t} \left\langle \nabla_{\Sigma} v \left(\frac{x}{\sqrt{t}} \right), \frac{1}{\sqrt{t}} \frac{\partial x}{\partial t} - \frac{x}{2t\sqrt{t}} \right\rangle \quad (39)$$

$$+ \frac{1}{\sqrt{t}} \Delta_{\Sigma} v \left(\frac{x}{\sqrt{t}} \right). \quad (40)$$

By the backwards MCF

$$\partial_t x = H \mathbf{n} = \frac{\langle x, \mathbf{n} \rangle}{2t} \mathbf{n},$$

we have

$$\frac{dw}{dt} + \Delta_{\Sigma_t} w = \frac{1}{\sqrt{t}} \left(\Delta_{\Sigma} v \left(\frac{x}{\sqrt{t}} \right) - \frac{1}{2} \left\langle \frac{x}{\sqrt{t}}, \nabla_{\Sigma} v \left(\frac{x}{\sqrt{t}} \right) \right\rangle + \frac{1}{2} v \left(\frac{x}{\sqrt{t}} \right) \right).$$

From [Lemma 3.4](#) and [Lemma 3.6](#), we have the following estimate:

Lemma 3.8. *Given $\epsilon_1 > 0$, there exists $R > R_3$ such that on $\bigcup_{t \in (0,1]} \Sigma_t \setminus B_R \times \{t\}$,*

$$\left| \frac{dw}{dt} + \Delta_{\Sigma_t} w \right| \leq \epsilon_1 (|w| + |\nabla_{\Sigma_t} w|).$$

Note that this is an analogue of the first part of [Lemma 2.12](#).

3.4 Proof of the Main Result: Carleman Estimates

From [Lemma 3.8](#) and [Lemma 3.6](#), we see that w is asymptotic to 0 as $t \rightarrow 0^+$. Wang modifies the proof E-S-Š's backwards uniqueness on parabolic equations to show the uniqueness result required for L_Σ . While most of the steps in the proof closely follow that of E-S-Š's, in the following we only introduce two key lemmas which directly imply [Theorem 1.1](#).

Lemma 3.9 (A key Carleman estimate). *Let $\alpha > \alpha_0$, $0 < T \leq 1$, $R > R_4$ and $Q_{R,T} = \{(x, t) : x \in \Sigma_t \setminus \bar{B}_R, t \in (0, T]\}$. Denote*

$$\mathcal{G}_{\alpha,T,R}(x, t) = \exp \left[2\alpha(T-t)(|x|^{1+\delta} - R^{1+\delta}) + 2|x|^2 \right].$$

Then for all $\phi \in C_c^\infty(Q_{R,T})$,

$$\begin{aligned} \int_0^T \int_{\Sigma_t} \left(|\phi|^2 + C_5 |\nabla_{\Sigma_t} \phi|^2 \right) \mathcal{G}_{\alpha,T,R} d\mu_t dt &\leq \int_0^T \int_{\Sigma_t} \left(\frac{d\phi}{dt} + \Delta_{\Sigma_t} \phi \right)^2 \mathcal{G}_{\alpha,T,R} d\mu_t dt \\ &\quad + \int_{\Sigma_T} |\nabla_{\Sigma_T} \phi|^2 \mathcal{G}_{\alpha,T,R} d\mu_T. \end{aligned} \quad (41)$$

Proof. Omitted. Escauriaza's paper discusses a simpler version of above estimate. $\square$

Lemma 3.10. *There exist $R_5 > R_3$ and $M > 0$ such that for $x \in \Sigma_t \setminus B_{R_5}$ and $t \in (0, 1]$,*

$$|w(x, t)| + |\nabla_{\Sigma_t} w(x, t)| \leq \exp \left(-\frac{M|x|^2}{t} \right). \quad (42)$$

Proof. Key ingredients: backwards MCF + [Lemma 3.4](#) + Carleman estimate (41). $\square$

Now we present the proof of our main theorem.

Proof of Theorem 1.1. The argument closely follows backwards uniqueness theorem's case.

- Choice of constants: Choose ϵ_1 in [Lemma 3.8](#) to be $\frac{1+C_5}{4}$; choose $R > 0$ sufficiently large such that all the previous lemmas can be applied; choose $T = M/16$.
- Choice of cutoffs: Given $a \in (0, 1)$ and $r \gg 1$, we consider $\psi_{a,r} \in C_c^\infty(\mathbb{R}^{n+1})$

$$\psi_{a,r}(x) = \begin{cases} 0, & |x| \leq (1+a)R \text{ or } |x| \geq 2r \\ 1, & (1+2a)R \leq |x| \leq r \end{cases}$$

and $|D\psi_{a,r}|, |D^2\psi_{a,r}|$ bounded above by a function of a and R .

We also choose a cutoff function η in time: given $0 < \epsilon < T/8$, $\eta_\epsilon \in C_c^\infty(\mathbb{R})$ satisfies

$$\eta_\epsilon(t) = \begin{cases} = 0, & t \leq \epsilon \\ = 1, & t \geq 2\epsilon \end{cases}$$

$0 \leq \eta_\epsilon \leq 1$ and $|\eta'_\epsilon| \leq 2\epsilon^{-1}$.

- Choose the test function $\phi_{\epsilon,a,r} = \eta_\epsilon \psi_{a,r} w$, we have

$$\begin{aligned} \frac{d}{dt} \phi_{\epsilon,a,r} + \Delta_{\Sigma_t} \phi_{\epsilon,a,r} &= \eta_\epsilon \psi_{a,r} \left(\frac{dw}{dt} + \Delta_{\Sigma_t} w \right) + 2 \langle \nabla_{\Sigma_t} \psi_{a,r}, \nabla_{\Sigma_t} w \rangle \eta_\epsilon \\ &\quad + (\langle D\psi_{a,r}, \partial_t x \rangle + \Delta_{\Sigma_t} \psi_{a,r}) \eta_\epsilon w + \eta'_\epsilon \psi_{a,r} w. \end{aligned} \quad (43)$$

By [Lemma 3.8](#),

$$\begin{aligned} \left| \frac{d}{dt} \phi_{\epsilon,a,r} + \Delta_{\Sigma_t} \phi_{\epsilon,a,r} \right|^2 &\leq \frac{1}{2} \left(|\phi_{\epsilon,a,r}|^2 + C_5 |\nabla_{\Sigma_t} \phi_{\epsilon,a,r}| \right) \\ &\quad + \lambda \left(|w|^2 + |\nabla_{\Sigma_t} w|^2 \right) \chi_{a,R,r} + 16\epsilon^{-2} |w|^2 \tilde{\chi}_{\epsilon,R,r}, \end{aligned} \quad (44)$$

where $\lambda = \lambda(n, C_1, a, R)$; $\chi_{a,R,r}$ is the characteristic function supported in $(B_{2r} \setminus B_r) \cup (B_{(1+2a)R} \setminus B_{(1+a)R})$; and $\tilde{\chi}_{\epsilon,R,r}$ is the characteristic function supported in $(B_{2r} \setminus B_R) \times (\epsilon, 2\epsilon)$.

Hence, applying [Lemma 3.9](#) and (44), we have, for $\alpha > \alpha_0$,

$$\begin{aligned} \int_0^T \int_{\Sigma_t} |\phi_{\epsilon,a,r}|^2 \mathcal{G}_{\alpha,T,R} d\mu_t dt &\lesssim_{n,\lambda,a,R,T} \int_0^T \int_{\Sigma_t \cap B_{(1+2a)R} \setminus B_{(1+a)R}} (|w|^2 + |\nabla_{\Sigma_t} w|^2) \mathcal{G}_{\alpha,T,R} d\mu_t dt \\ &\quad + \int_0^T \int_{\Sigma_t \cap B_{2r} \setminus B_R} (|w|^2 + |\nabla_{\Sigma_t} w|^2) \mathcal{G}_{\alpha,T,R} d\mu_t dt \\ &\quad + \epsilon^{-2} \int_\epsilon^{2\epsilon} \int_{\Sigma_t \cap B_{2r} \setminus B_R} |w|^2 \mathcal{G}_{\alpha,T,R} d\mu_t dt + \int_{\Sigma_T} |\nabla_{\Sigma_T} \phi_{\epsilon,a,r}|^2 \mathcal{G}_{\alpha,T,R} d\mu_T. \end{aligned} \quad (45)$$

- By the asymptotically conical structure of Σ_t , we know that there exists $V_0 > 0$, depending only on the geometry of C , such that if t is sufficiently small, then $\text{Vol}(\Sigma_t \cap B_{2r} \setminus B_R) \leq V_0 r^n$. Also, it follows from [Lemma 3.6](#) (bounds on $|\nabla_{\Sigma_t}^i v|$) that $|w(x, t)| \leq C_3 t$ for $x \in \Sigma_t \cap B_{2r} \setminus B_R$. Thus, letting $\epsilon \rightarrow 0$, we see that the integral with ϵ^{-2} term in (45) converges to 0.

- By MCT, we know that

$$\begin{aligned} \int_0^T \int_{\Sigma_t} |\psi_{a,r} w|^2 \mathcal{G}_{\alpha,T,R} d\mu_t dt &\lesssim \int_0^T \int_{\Sigma_t \cap B_{(1+2a)R} \setminus B_{(1+a)R}} (|w|^2 + |\nabla_{\Sigma_t} w|^2) \mathcal{G}_{\alpha,T,R} d\mu_t dt \\ &\quad + \int_0^T \int_{\Sigma_t \cap B_{2r} \setminus B_R} (|w|^2 + |\nabla_{\Sigma_t} w|^2) \mathcal{G}_{\alpha,T,R} d\mu_t dt \\ &\quad + \int_{\Sigma_T} |\nabla_{\Sigma_T} (\psi_{a,r} w)|^2 \mathcal{G}_{\alpha,T,R} d\mu_T \end{aligned} \quad (46)$$

Furthermore, it follows from the fact that $M/T > 8$, [Lemma 3.10](#) and (46) yields

$$\begin{aligned} & \exp \left\{ [(1 + 17a)^{1+\delta} - 1] \alpha T R^{1+\delta} \right\} \int_0^{\frac{T}{2}} \int_{\Sigma_t \cap B_r \setminus B_{(1+17a)R}} w^2 d\mu_t dt \\ & \lesssim r^2 \exp \left(2^{2+\delta} \alpha T r^{1+\delta} - \frac{Mr^2}{2T} \right) \exp \left\{ 2[(1 + 2a)^{1+\delta} - 1] \alpha T R^{1+\delta} \right\} + 1. \end{aligned} \quad (47)$$

- Note that $(1 + 17a)^{1+\delta} - 1 \geq 17a$ and $(1 + 2a)^{1+\delta} - 1 \leq 8a$. Thus, letting $r \rightarrow \infty$,

$$\int_0^{T/2} \int_{\Sigma_t \setminus B_{(1+17a)R}} w^2 d\mu_t dt \lesssim \exp(-\alpha a T R^{1+\delta}). \quad (48)$$

Letting $\alpha \rightarrow \infty$ and by the arbitrariness of $a > 0$, we can conclude that $w \equiv 0$ on $Q_{R,T/2}$ and thus $v \equiv 0$ on $\Sigma \setminus B_{2R/\sqrt{T}}$.

- There is also a version of unique continuation theorem with weaker assumptions. Consider the equation $Lu = -\operatorname{div}(A(x)\nabla u) + V(x)u = 0$, where we assume there exist two constants $\delta, C > 0$ such that, if $V(x) = V^+(x) - V^-(x)$ then

$$0 \leq V^+(x) \leq \frac{C}{|x - x_0|^2} \quad (49)$$

$$0 \leq V^-(x) \leq \frac{\delta}{|x - x_0|^2} \quad (50)$$

for any $x \in B_{r_0} \cap \Omega$.

Theorem 3.11 (Garofalo-Lin). *Let $u \in H_{loc}^{1,2}(\Omega)$ be a solution of $Lu = -\operatorname{div}(A(x)\nabla u) + V(x)u = 0$ in Ω , where V satisfies the assumptions (49), (50). If at a point $x_0 \in \Omega$ we have*

$$\int_{|x-x_0|<r} u^2 dx = O(\exp(-Ar^{-\alpha})) \quad \text{for some } A, \alpha > 0,$$

as $r \rightarrow 0$, then $u \equiv 0$ in Ω .

Applying the above uniqueness continuation theorem, with the bounds on the function Q and the expression of the elliptic operator L_Σ , we conclude that $L_\Sigma v \equiv 0$ on the entire Σ .

□